

$$\chi^2$$

$$p > n$$

E	
Θ	θ
L_n	θ

$$E$$

$$\Theta$$

$$\{P_\theta\}\theta$$

$$\theta\\(E,\{P_\theta\}_{\theta\in\Theta})$$

$$EX$$

$$\Theta\theta\Theta\subseteq\mathbb{R}^k$$

$$P_\theta\theta$$

$$E=\{0,1\}$$

$$\Theta=[0,1]p$$

$$\{P_\theta\}(p)$$

$$P\{P_\theta\}$$

$$P$$

$$\theta \mapsto P_\theta$$

$$\theta \neq \theta' \implies P_\theta \neq P_{\theta'}$$

$$\theta P_\theta \theta$$

$$X_{\mu_1+\mu_2}$$

$$X\sim\mathcal{N}(\mu_1+\mu_2,1)$$

$$\mu_1=2,\mu_2=3\rightarrow$$

$$\mu_1=1,\mu_2=4\rightarrow$$

$$\mu_1\mu_2(\mu_1,\mu_2)$$

$$\theta x$$

$$x\theta$$

$$X_1,...,X_nL_n(\theta)$$

$$L_n(\theta)=\prod_{i=1}^nf(X_i;\theta)$$

$$X_i\theta$$

$$pX_1=1,X_2=1,X_3=0$$

$$p=0.5$$

$$L_3(0.5)=0.5\times 0.5\times (1-0.5)=0.125$$

$$p=0.9$$

$$L_3(0.9)=0.9\times 0.9\times (1-0.9)=0.081$$

$$p=0.5p=0.5$$

$$L_n\Pi n\Sigma$$

$$\ell_n(\theta)=\sum_{i=1}^nf(X_i;\theta)$$

$$Tf(t;\lambda)=\lambda e^{-\lambda t}$$

$$x_1=2, x_2=3, x_3=5$$

$$L(\lambda) = (\lambda e^{-2\lambda}) \times (\lambda e^{-3\lambda}) \times (\lambda e^{-5\lambda}) = \lambda^3 e^{-10\lambda}$$

$$\ell(\lambda) = (\lambda^3 e^{-10\lambda}) = 3\,\lambda - 10\lambda$$

$$\lambda$$

$$\theta \!\rightarrow\! x$$

$$x\!\rightarrow\!\theta$$

$$\theta$$

$$\theta$$

$$(E,\{P_\theta\})$$

$$\theta \neq \theta' \implies P_\theta \neq P_{\theta'}$$

$$X\theta$$

$$\sum f$$

$$\theta$$

$$\theta \hat{\theta}$$

$\hat{\theta}_n$	

$$\theta$$

$$\hat{\theta}$$

$$\hat{\theta}$$

$$\hat{\theta}_n X_1, \ldots, X_n$$

$$\hat{\theta}_n = g(X_1, \ldots, X_n)$$

$$g\theta$$

$$\theta$$

$$\ell_n(\theta)$$

$$\ell_n(\theta) = \sum_{i=1}^n f(X_i;\theta)$$

$$\theta \frac{\partial}{\partial \theta} \ell_n(\theta)$$

$$X(\lambda)\theta=\lambda$$

$$P(X=x)=\frac{e^{-\lambda}\lambda^x}{x!}$$

$$X_1,...,X_n$$

$$\ell_n(\lambda) = \sum (-\lambda + X_i \, \lambda - \, X_i!) = -n\lambda + (\sum X_i) \, \lambda - C$$

$$\frac{\partial}{\partial \lambda} \ell_n(\lambda) = -n + \frac{\sum X_i}{\lambda}$$

$$-n + \frac{\sum X_i}{\hat{\lambda}} = 0 \implies \hat{\lambda}_{MLE} = \tfrac{1}{n} \sum X_i = \bar{X}$$

$$\theta E[X]$$

$$\bar{X}$$

$$\hat{\theta} = 2$$

$$km_k(\theta) = \mathbb{E}[X^k] \tfrac{1}{n} \sum X_i^k$$

$$\mathbb{E}[X] = \bar{X}()$$

$$E[X^2]$$

$$(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

$$\theta$$

$$Bias = 0$$

$$Bias \neq 0$$

$$(\hat{\theta})=\mathbb{E}[(\hat{\theta}-\mathbb{E}[\hat{\theta}])^2]$$

$$(\hat{\theta})=\mathbb{E}[(\hat{\theta}-\theta)^2]=\underbrace{(\hat{\theta})^2}+(\hat{\theta})$$

$$pn=100, \sum x_i=1$$

$$X\sim (p)$$

$$\hat{p}_{MLE}=\frac{1}{100}=0.01\;(1)$$

$$\hat{p}=0$$

$$n\rightarrow \infty n$$

$$\hat{\theta}=0.01$$

$$\nabla \ell = 0$$

$$=^2+$$

$$\hat{\theta}nn \rightarrow \infty$$

$$n$$

I	
	$\hat{\theta}f(\hat{\theta})$

$$n$$

$$n \rightarrow \infty \theta^*$$

$$\hat{\theta}_n \theta^*$$

$$\forall \epsilon > 0,_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta^*| > \epsilon) = 0$$

$$\epsilon$$

$$n\hat{\theta}_n\theta^*\rightarrow 0$$

$$\hat{\theta}_n-\theta^*\rightarrow 0$$

$$\sqrt{n}\sqrt{n}(\hat{\theta}_n-\theta^*)\rightarrow$$

$$\sqrt{n}(\hat{\theta}_n - \theta^*) \overset{d}{\rightarrow} \mathcal{N}(0, \sigma_{asymp}^2)$$

$$\sigma_{asymp}^2$$

$$\rightarrow$$

$$\rightarrow$$

$$\theta$$

$$I(\theta)=-\mathbb{E}\left[\frac{\partial^2}{\partial \theta^2}\;f(X;\theta)\right]$$

$$f(x;p)=p^x(1-p)^{1-x}$$

$$f=x\;p+(1-x)\;(1-p)$$

$$\frac{x}{p}-\frac{1-x}{1-p}$$

$$-\frac{x}{p^2}-\frac{1-x}{(1-p)^2}$$

$$E[X]=pI(p)=\frac{p}{p^2}+\frac{1-p}{(1-p)^2}=\frac{1}{p}+\frac{1}{1-p}=\frac{1}{p(1-p)}$$

$$\frac{1}{nI(p)}=\frac{p(1-p)}{n}\bar{X}\bar{X}$$

$$\hat{\theta} g(\hat{\theta})$$

$$g(\theta)$$

$$\sqrt{n}(g(\hat{\theta}_n)-g(\theta))\overset{d}{\rightarrow}\mathcal{N}(0,[g'(\theta)]^2\cdot\sigma^2)$$

$$\sigma^2[g'(\theta)]^2$$

$$p$$

$$n=10,000 \hat{p}=0.6$$

$$p\frac{p}{1-p}$$

$$\hat{p}\frac{\hat{p}}{1-\hat{p}}$$

$$g(p)=\frac{p}{1-p}$$

$$g'(p)=\frac{1}{(1-p)^2}$$

$$\frac{p(1-p)}{n}[g'(p)]^2$$

$$n\leq n\leq 30n$$

$$\sqrt{n}n$$

$$\hat{\theta}-\theta\frac{1}{\sqrt{n}}$$

$$n\infty$$

$$\sqrt{n}$$

$$\hat{\theta}$$

$$n\rightarrow\infty$$

$$\sqrt{n}\mathcal{N}(0,I(\theta)^{-1})$$

$$g(\hat{\theta})g'(\theta)^2\times(\hat{\theta})$$

$$\hat{\theta}$$

$1-\alpha$	0.95
α	
$q_{\alpha/2}$	
	θ
	n

$$\theta$$

$$\theta$$

$$[0.4, 0.6] \rightarrow$$

$$\rightarrow$$

$$\hat{\theta}\theta$$

$$\hat{\theta}_n$$

$$\sqrt{n}(\hat{\theta}_n - \theta) \overset{d}{\rightarrow} \mathcal{N}\left(0, \frac{1}{I(\theta)}\right)$$

$$Z \sim \mathcal{N}(0,1)$$

$$\sqrt{nI(\theta)}(\hat{\theta}_n-\theta) \approx Z$$

$$[-1.96,1.96]q_{0.025} \approx 1.96$$

$$P\left(-q_{\alpha/2} \leq \sqrt{nI(\theta)}(\hat{\theta}_n-\theta) \leq q_{\alpha/2}\right) \approx 1-\alpha$$

$$I(\theta)\theta I(\theta)I(\hat{\theta}_n)$$

$$\hat{\theta}_n \rightarrow \theta I(\hat{\theta}_n) \rightarrow I(\theta)$$

$$n$$

$$\theta$$

$$\mathcal{I}=\left[\hat{\theta}_n-\frac{q_{\alpha/2}}{\sqrt{nI(\hat{\theta}_n)}},\hat{\theta}_n+\frac{q_{\alpha/2}}{\sqrt{nI(\hat{\theta}_n)}}\right]$$

$$\hat{\theta}_n$$

$$\frac{1}{\sqrt{n}}\!\times\!\frac{1}{\sqrt{I}}$$

$$I(\theta)$$

$$pp(1-p)$$

$$p\hat{p}$$

$$p(1-p)p=0.50.25$$

$$0.25$$

$$\hat{p} \pm 1.96 \frac{\sqrt{0.25}}{\sqrt{n}} = \hat{p} \pm \frac{0.98}{\sqrt{n}} \approx \hat{p} \pm \frac{1}{\sqrt{n}}$$

$$\pm 3.1n=1000\!\approx\!3.1$$

$$\hat{p}_A=0.05$$

$$\hat{p}_B=0.07$$

$$\Delta=0.02$$

$$\approx \sqrt{\frac{p(1-p)}{n}+\frac{p(1-p)}{n}}$$

$$\approx \sqrt{\frac{0.05\times 0.95}{1000}+\frac{0.07\times 0.93}{1000}}\approx 0.011$$

$$95=[0.02-1.96(0.011),0.02+1.96(0.011)]=[-0.001,0.041]$$

$$[-0.1,4.1]$$

$$n\rightarrow\rightarrow$$

$$\rightarrow$$

$$\theta\theta\hat{\theta}$$

$$[0.4,0.6]$$

$$\theta$$

$$\hat{\theta} \pm 1.96 \times$$

$$\theta\hat{\theta}$$

$$H_0H_1$$

$$\alpha 1-\beta$$

$$\hat{p} = 0.6$$

$$H_0, H_1 \alpha$$

H_0	
H_1	
α	
β	
$1 - \beta$	
	H_0

$$H_1$$

$$H_0$$

$$\psi H_0 H_0 H_0$$

$$\Theta \Theta_0 \Theta_1$$

$$\psi(X)$$

$$\psi(X)=\begin{cases} 1H_0H_1\\ 0H_0H_0 \end{cases}$$

$\backslash$	H_0	H_0
H_0	$\checkmark$	α
H_1	β	$1-\beta$

$$\alpha$$

$$\alpha \leq 0.05$$

$$\psi$$

$$P_{H_0}(\psi(X)=1) \leq \alpha \alpha = 0.05$$

$$P_{H_1}(\psi(X)=1)$$

$$H_1, p>0.5H_0:p=0.5X$$

$$X \geq k$$

$$\alpha H_0(p=0.5)$$

$$P(X \geq 9) = P(9) + P(10) \approx 0.0098 + 0.0010 = 0.0108 < 0.05$$

$$P(X \geq 8) = P(8) + \cdots \approx 0.0439 + 0.0108 = 0.0547 > 0.05$$

$$k$$

$$p=0.8=P_{p=0.8}(X\geq 9)\approx 0.37p=0.8n$$

$$H_0$$

$$p=P(T(X)\geq t_{obs} \mid H_0)$$

$$t_{obs}H_0$$

$$<0.05H_0\rightarrow H_0$$

$$>0.05H_0\rightarrow H_0$$

$$\mu_0=1\bar{X}=1.05\sigma=0.2$$

$$H_0\mu=1$$

$$H_1\mu>1$$

$$Z=\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}=\frac{1.05-1}{0.2/\sqrt{30}}\approx\frac{0.05}{0.0365}\approx1.37$$

$$Z\geq 1.370.085$$

$$p(0.085)>\alpha(0.05)$$

$$H_0$$

$$nn\rightarrow\infty$$

$$H_0H_1$$

$$\alpha\beta$$

$$\alpha 1-\beta$$

$$H_0\alpha H_0$$

$$\alpha, \beta$$

$$\alpha$$

λ	$H_0H_1H_0$
	χ^2
	H_0
	H_0

$$\hspace*{10cm}$$

$$\theta_0\theta_1$$

$$H_0$$

$$H_1$$

$$L(\theta_0)L(\theta_1)$$

$$\frac{L(\theta_1)}{L(\theta_0)} > k(H_1)$$

$$\hspace*{10cm}$$

$$\lambda = \frac{\theta \in \Theta_0}{\theta \in \Theta} \frac{L(\theta)}{L(\theta)} = \frac{L(\hat{\theta}_{H_0})}{L(\hat{\theta}_{MLE})}$$

$$H_0$$

$$\lambda H_0H_0$$

$$n\rightarrow\infty$$

$$-2\,\lambda\stackrel{d}{\rightarrow}\chi_d^2$$

$$dH_0H_1d=1$$

$$H_0$$

$$\hat{\theta}_{MLE}$$

$$\theta_0$$

$$L(\hat{\theta})L(\theta_0)$$

$$2[\ell(\hat{\theta})-\ell(\theta_0)]$$

$$\hat{\theta}\hat{\theta}-\theta_0$$

$$\theta$$

$$\frac{(\hat{\theta}-\theta_0)^2}{(\hat{\theta})}$$

$$\theta_0$$

$$\approx 0H_0H_0$$

$$H_0$$

$$\sigma^2S^2$$

$$X_i\sim\mathcal{N}(\mu,\sigma^2)\mu\sigma$$

$$Z\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}\rightarrow$$

$$t\frac{\bar{X}-\mu_0}{S/\sqrt{n}}\rightarrow$$

$$n\rightarrow\infty S\rightarrow\sigma$$

$$pH_0:p=0.5H_1:p\neq 0.5n=100\hat{p}=0.6$$

$$|\hat{p}-p_0|=|0.6-0.5|=0.1$$

$$H_0\sqrt{\frac{p_0(1-p_0)}{n}}=\sqrt{\frac{0.5\times 0.5}{100}}=0.05$$

$$Z$$

$$W=\;=\frac{0.1}{0.05}=2.0$$

$$|W|=2.0>1.96$$

$$H_0$$

$$\hspace*{10cm}\hrule\hspace*{10cm}$$

$$n\rightarrow\infty$$

$$\approx$$

$$e^{-x^2/2}\!-\!x^2/2x^2\!-\!2$$

$$\theta$$

$$-2\,()\sim\chi^2$$

$$\hat{\theta}$$

$$\hat{\theta}$$

$$\theta$$

$$\mathcal{P}$$

$$\chi^2$$

O_j	N_j
E_j	np_j
χ^2	$\sum \frac{(O-E)^2}{E}$
	$K-1-d$

$$K$$

$$\rightarrow K=4$$

$$\rightarrow K=4$$

$$N_1,...,N_K$$

$$\chi^2$$

$$(O-E)^2E$$

$$T_n$$

$$T_n = \sum_{j=1}^K \frac{(N_j - np_j^0)^2}{np_j^0} = \sum_{j=1}^K \frac{(-)^2}{}$$

$$(O-E)^2$$

$$E$$

$$\approx Z\chi^2$$

$$n\rightarrow\infty$$

$$T_n\overset{d}{\rightarrow}\chi^2_{K-1}$$

$$KK-1n\sum N_j=nN_KK-1$$

$$\hat{\theta}$$

$$\lambda$$

$$\hat{\lambda}$$

$$\hat{\lambda} p_j(\hat{\lambda})$$

$$T_n = \sum \frac{(N_j - np_j(\hat{\theta}))^2}{np_j(\hat{\theta})} \overset{d}{\rightarrow} \chi^2_{K-1-d}$$

$$d$$

$$dO-Edf$$

$$K=3$$

$$n=100$$

$$O$$

$$E$$

$$100\times0.1=10$$

$$100\times0.3=30$$

$$100\times0.6=60$$

$$\begin{aligned}\chi^2 &= \frac{(5-10)^2}{10} + \frac{(25-30)^2}{30} + \frac{(70-60)^2}{60} \\ &= \frac{25}{10} + \frac{25}{30} + \frac{100}{60} = 2.5 + 0.83 + 1.67 = 5.0\end{aligned}$$

$$K-1=2\chi^2_2$$

$$5.0 < 5.99()$$

$$E_jE\geq 5$$

$$(O-E)\;(O-E)E$$

$$\theta X \rightarrow Y$$

$$\sum \frac{(O-E)^2}{E}$$

$$K-1K-1-()$$

$$p\mu XY$$

$$XYY=X\beta$$

	$n\times p$
β	
$e\hat{e}$	$Y\hat{Y}$
P	YX

$$\begin{array}{c} \hline \end{array}$$

$$=\beta+\epsilon$$

$$\beta$$

$$\epsilon$$

$$=\beta+\epsilon$$

$$n\times pnp$$

$$\mathbb{E}[\epsilon]=0$$

$$(\epsilon)=\sigma^2I_n$$

$$\begin{array}{c} \hline \end{array}$$

$$\beta$$

$$\|Y-\beta\|^2\hat{\beta}$$

$$\hat{\beta} =_{\beta} \|-\beta\|^2$$

$$\beta$$

$${}^T\beta = {}^T$$

$$({}^T)$$

$$\hat{\beta} = ({}^T)^{-1T}$$

$$\hat{Y}Y$$

$$n$$

$$\mathcal{C}()n$$

$$Y\mathcal{C}()e=Y-\hat{Y}\mathcal{C}()$$

$${}^Te=0\implies{}^T(Y-\hat{\beta})=0$$

$${}^T\beta = {}^T$$

$$\hat{Y}YP$$

$$\hat{Y}=PY,P=(^T)^{-1T}$$

$$PY\hat{Y}$$

$$\hat{\beta}$$

$$Y$$

$$\beta \mathbb{E}[\hat{\beta}] = \beta$$

$$\epsilon \sim \mathcal{N}(0,\sigma^2I)$$

$$\hat{\beta}Y$$

$$\hat{\beta} \sim \mathcal{N}\left(\beta,\sigma^2({}^T)^{-1}\right)$$

$$\beta_3H_0:\beta_3=0$$

$$T=\frac{\hat{\beta}_j-0}{(\hat{\beta}_j)}\sim t_{n-p}$$

$$(\hat{\beta}_j)\hat{\sigma}\sqrt{({}^{(T)})_{jj}^{-1}}$$

$$\hat{\beta}\hat{\epsilon}\hat{Y}e$$

$$Y$$

$$X_1$$

$$X_2$$

$$X_3$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

$$\hat{\beta} = [100, 2.5, 0.1, 50]^T$$

$$X_2$$

$$\rightarrow$$

$$^{(T)}$$

$$L^2$$

$$Y$$

$$Y = \beta + \epsilon$$

$$\hat{\beta} = (^T)^{-1T}Y$$

$$\hat{Y}Y\mathcal{C}()$$

$$Y = X\beta Y$$

$$X\beta$$

g	$X\beta\mu$
	$\pm\infty$
	∞

$$f(y) = h(y)e^{\eta T(y) - A(\eta)}$$

$$A'(\eta) = , A''(\eta) =$$

$$f(y;\theta) = h(y) \left(\eta(\theta) \cdot T(y) - A(\theta) \right)$$

$$\eta$$

$$T(y)$$

$$A(\theta)$$

$$X\beta-\infty \sim \infty\mu$$

$$Y \sim (p)$$

$$X\beta = 100$$

$$p$$

$$g(p) = \left(\frac{p}{1-p}\right) = X\beta$$

$$p = \frac{e^{X\beta}}{1+e^{X\beta}} = \frac{1}{1+e^{-X\beta}}$$

$$Y \in \{0,1,2,\ldots\}$$

$$\lambda$$

$$g(\lambda) = (\lambda) = X\beta \implies \lambda = e^{X\beta}$$

$$XY$$

$$\beta$$

$$\beta e^{X\beta}$$

$$\beta$$

$$XY \in \{0,1\}$$

$$Y \sim (p)(\frac{p}{1-p}) = \beta_0 + \beta_1 X$$

$$\hat{\beta}_0 = -10, \hat{\beta}_1 = 3$$

$$\eta = -10 + 3(3.8) = -10 + 11.4 = 1.4$$

$$p = \frac{1}{1+e^{-1.4}} \approx \frac{1}{1+0.246} \approx 0.80$$

$$\beta_1 = 3e^3 \approx 20$$

$$X\beta \; P(Y=1)=\beta_0+\beta_1X X(-\infty,\infty)X\beta$$

$$\rightarrow Y=X\beta$$

$$\rightarrow$$

$$\rightarrow$$

$$YY$$

$$g(\mu)=X\beta$$

$$g=$$

$$g=$$

$$\mu f$$

$$fh$$

h	

$$X_iK$$

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-X_i}{h}\right)$$

$$K(\cdot)\int K=1$$

$$h$$

$$\begin{array}{l} hh \\ \hat{f}f \\ h \end{array} = {}^2 +$$

$$h$$

$$\begin{array}{l} hn \\ nh \\ 1/\sqrt{nn}^{-2/5} \end{array} \qquad h_{opt} \propto n^{-1/5}$$

$$\mu$$

$$\begin{array}{l} h \\ h \\ h \end{array}$$

$$K\setminus h$$

$$\mu\sigma$$

$$\mathcal{N}f$$

$$Kh$$

$$h\uparrow\uparrow$$

$$hn^{-1/5}$$

Y

p

v	
λ	

$Var = 0$

$Var > 0$

xy

$$Av = \lambda v$$

$$S$$

$$S = \frac{1}{n-1}T$$

S

$$Sv_j = \lambda_j v_j$$

v_j

λ_j

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$$

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^p \lambda_j} \geq 0.95()$$

k

$$kp \rightarrow k$$

→

→

$$0.7 \times + 0.3 \times$$

S

$$p$$

$$p \gg n$$

$$p > n$$

$$Y=X\beta+\epsilon np$$

$$pnp\gg n$$

$p\gg n$	

$$p>n$$

$$\hat{\beta}=(X^TX)^{-1}X^TYp>nX^TX$$

$$x+y+z=10(1,1,8),(2,3,5)...$$

$$Y\beta_j\neq 0s$$

$$\beta^*psns\ll n$$

$$\|\beta^*\|_0:=\sum_{j=1}^p\mathbb{I}(\beta_j^*\neq 0)=s\ll n$$

$$ps$$

$$\hat{\beta} =_{\beta} \left(\|Y - X\beta\|^2 + \lambda \cdot (\beta) \right)$$

$$\lambda \lambda$$

$$=\sum \beta_j^2$$

$$=\sum |\beta_j|$$

$$\lambda$$

$$\beta_1^2+\beta_2^2\leq C\beta_1\neq 0,\beta_2\neq 0$$

$$|\beta_1|+|\beta_2|\leq C\beta_1=C,\beta_2=0\beta_2$$

$$ps$$

$$ppp\gg n$$

$$\|\hat{\beta}_{LASSO}-\beta^*\|^2\leq C\frac{s\,p}{n}$$

$$n=500$$

$$p=10,000$$

$$p\gg n$$

$$\rightarrow$$

$$\lambda \rightarrow \lambda \lambda$$

$$p \gg n$$

$$s$$

$$s \rightarrow p/n$$